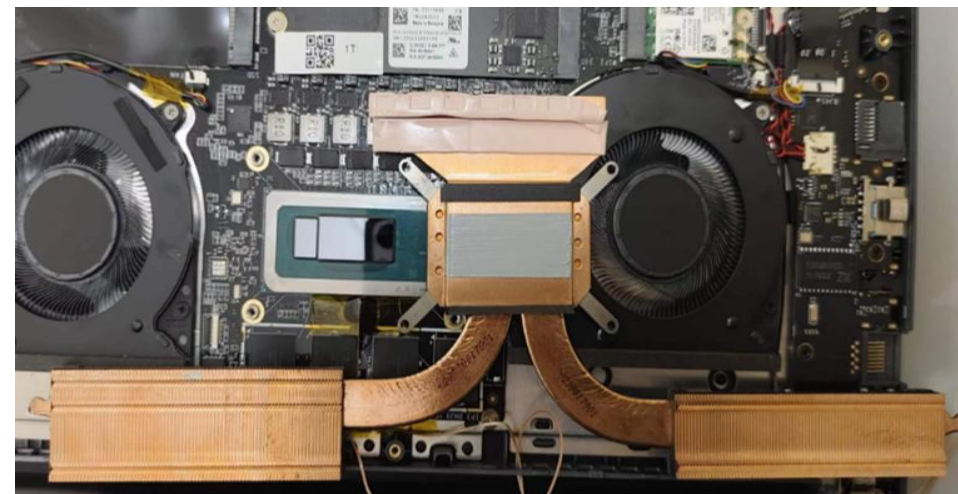
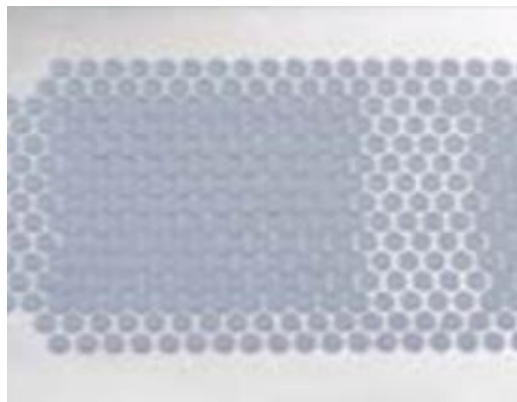
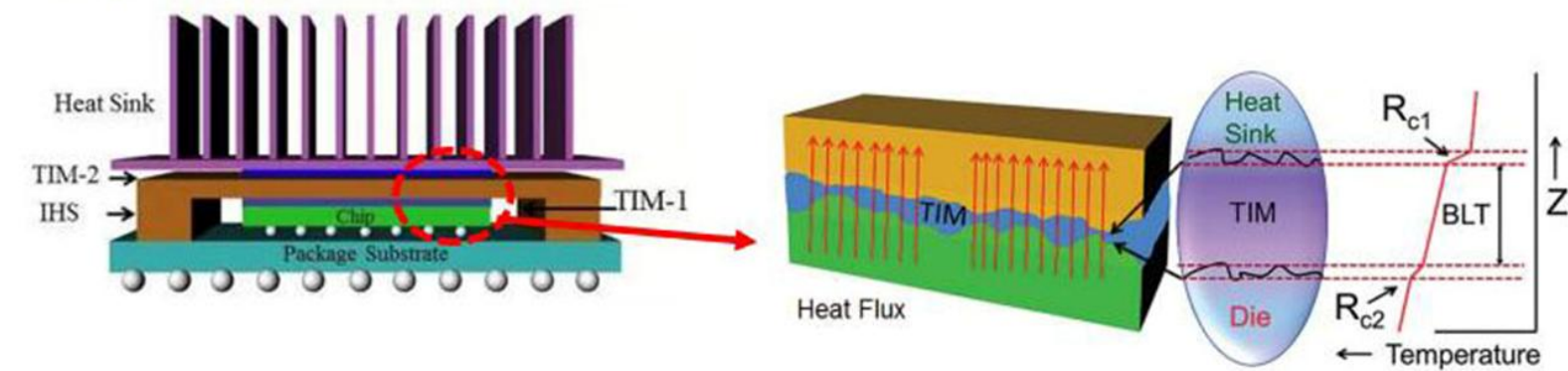


Deep-Materials ELM, FlexiMetal TIMs



Use fundamental science to make practical improvements in TIM

Deep-Materials
TIM Product Brochure

Use *fundamental science* to make practical improvements in Thermal Materials

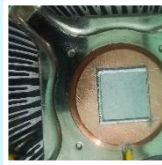
CPU/GPU TIMs



Grease

- Best in class performance
- Significantly reduce pump out

Highest performance traditional grease & Innovative Liquid Metal grease with low resistance



FlexiMetal

- Multi-phase Liquid metal – high viscosity
- Practical to use
- Highest performance

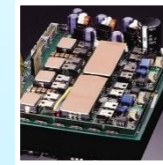
Brand new category of TIM with perf. of LM but ease of grease



Gels

- Highest conductivity
- Almost no oil seepage

Many kinds of unique gels – e.g. ultra low oil seepage gel, low dielectric constant gels



Gap Pads

- Almost eliminate oil seepage
- Reliable to very high temperatures

Ultra low oil seepage & ultra low outgassing silicone pads



CPU/GPU/Low BLT TIMs

- For Applications with mechanical attach
- High power AI Chips, GPUs, DPUs, CPUs, Networking,
- Performance measured in thermal resistance ($^{\circ}\text{C}\text{-cm}^2/\text{W}$)
- Our Offerings in this space – Greases, Liquid Metals

ELM, FlexiMetal

Applications:
CPU, GPU, Laptops, Desktop, Gaming, Servers
Power Electronics, Cold plates, AI Chips

P/N	TCG-ELM-8000	TCG-ELM-10000	BLM™ (Bonded Liquid Metal)	Testing Method
Type of product	Grease	Grease	Soft Metal	
Color	Grey	Grey	Silvery	Visual
BLT (mm)	0.025	0.025	/	—
Density (g/cc)	5.1	5.4	7.0	ASTM D792
Viscosity (cps)	200,000	260,000	100,000	ASTM D2196
Operation Temperature (°C)	-40~150	-40~150	-40~225	—
Electrical Parameters				
Volume Resistance ($\Omega\cdot\text{cm}$)	10^2	10^2	/	ASTM D257
Breakdown Voltage (Kv/mm)	/	/	/	ASTM D149
Liquidus point (°C) for non-eutectic alloy	/	/	~300	
Solidus point (°C) for non-eutectic alloy	/	/	~30	
Thermal Performance				
Thermal Conductivity (W/m-K)	8.0	10.0	35	ASTM D5470
Thermal Resistance @40 psi (°C -cm ² /W)	0.02	0.018	0.01	ASTM D5470 -

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